

1. Coupling

1.1.Content coupling

Related modules	Description	Improvement Direction
None		

1.2.Common coupling

Related modules	Description	Improvement Direction
SecurityConfig	Using a static singleton instance WHITE_LIST, share with JwtFilter	Remove WHITE_LIST from JwtFilter and define a check method inside security config to verify the route is accepted or not
Regex	Using static variables to contain regular expressions to check inputs	Keep it the same as this is an utility class

1.3.Control coupling

Related modules	Description	Improvement Direction
none		

1.4.Stamp coupling

Related modules	Description	Improvement Direction
CartService	addToCart and checkProductExistIncart are taking the whole User as parameter, but only need user email	Passing only String email instead of User user.
JwtService	generateToken takes the whole User userDetails but only use role and username attribute	Passing only String userRole and String username instead of User userDetails
ProductService	getWishList are taking the whole User but only user email	Passing only String userMail instead

1.5.Data coupling

Related modules	Description	Improvement Direction
JwtFilter		
ModelMapper		
ProductHashMapConverter		

AuthController		
CartController		
ChatController		
OrderController		
ProductController		
ReviewController		
AimsOtpException		
CustomExceptionHandler		
GeneralException		
UserExistException		
LoginRequest		
RegisterRequest		
BaseResponse		
LoginResponse		
BaseEntity		
FavoriteProductUser		
Invoice		
Order		
OrderService		
OrderProduct		
Product		
ProductCart		
Review		
TransactionInfo		
User		
UserCart		
OrderRepository		
ProductCartRepository		
ProductRepository		
ReviewRepository		
UserRepository		
WishlistRepository		
EmailService		
MediaService		
OtpService		
UserService		

2. Cohesion

2.1.Coincidental cohesion

Related modules	Description	Improvement Direction
none		

2.2.Logical cohesion

Related modules	Description	Improvement Direction
None		

2.3.Temporal cohesion

Related modules	Description	Improvement Direction
None		

2.4.Procedural cohesion

Related modules	Description	Improvement Direction
none		

2.5.Communication cohesion

Related modules	Description	Improvement Direction
none		

2.6.Sequential cohesion

Related modules	Description	Improvement Direction
none		

2.7.Functional cohesion

Related modules	Description	Improvement Direction
JwtFilter	methods are related to managing and interacting with specific entities or functionalities, ensuring that the provided methods are logically grouped together based on their functionality	
ModelMapper		
SecurityConfig		
Regex		
AuthController		
CartController		
OrderController		
ProductController		
ReviewController		
AimsOtpException		
CustomExeptionHandler		
GeneralExeption		
UserExistException		
LoginRequest		
RegisterRequest		
BaseResponse		
LoginResponse		
BaseEntity		
FavoriteProductUser		

Invoice		
Order		
OrderProduct		
Product		
ProductCart		
Review		
TransactionInfo		
User		
UserCart		
OrderRepository		
ProductCartRepository		
ReviewRepository		
UserRepository		
WishListRepository		
CartService		
EmailService		
JwtService		
MediaService		
OrderService		
OtpService		
ProductService		
UserService		